

Mini Project - 3: Simplified CDO Analysis (Part 1)

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Interactive dashboard: <https://pawarraj8888.github.io/valuation-assignments/cdo-analysis/> (the same model in the browser: pick any of the 1000 cases or change the inputs, and the cash flows, distributions and sensitivities update)

Aim: To build a simulation model of a CDO backed by 10 speculative grade corporate bonds, with correlated default dates drawn from fixed random numbers, to distribute the pool's quarterly cash flows to Class A, Class B and the bank's equity through the waterfall, and to describe the resulting cash flows statistically, including their sensitivity to the main inputs. Valuation of the classes is Part 2.

1. Key Assumptions

Collateral: 10 bonds, \$10 MM face each, 5 years, 6% annual coupon paid quarterly. Each bond promises \$0.15 MM per quarter and \$10.15 MM in quarter 20, so the pool promises \$1.5 MM per quarter, \$101.5 MM at maturity and \$130 MM in total.

Default Model: We use the BIS debt model from class. $(1 - \text{LGD}) = 40\%$ of every promised payment is non-defaultable and is always paid. The other 60% stops in the quarter of default. So a defaulted bond still pays 40% of its coupons and 40% of its principal, on the original dates.

Default Timing: Each bond has a 4% annual default probability. Its default time in years comes from the formula in class, $t = \ln(1 - u) / \ln(1 - 0.04)$, and the bond defaults in quarter $\text{ceil}(4t)$. Past quarter 20 it survives.

Correlation: We apply the 0.20 correlation to the normal random numbers behind the default times (Cholesky factor of a 10×10 matrix with 0.20 off the diagonal) and then take $u = N(x)$.

Random Numbers: 1000 cases $\times$ 10 bonds of standard normals, generated once (seed 6103) and stored in fixed_random_numbers.csv. Cases are numbered 1 to 1000 and every run uses the same numbers. Before use they are de-meant and moment matched as shown in class, so that across the 1000 cases each bond's numbers have mean 0 and variance 1 and are uncorrelated with the other bonds' numbers.

Classes: Class A: \$20 MM, 2% coupon. Class B: \$10 MM, 4% coupon. We treat both as bullet bonds with quarterly coupons and principal at year 5, since the collateral only returns principal at maturity. Class A is owed \$0.10 MM per quarter and \$20.10 MM at maturity, Class B \$0.10 MM and \$10.10 MM.

Waterfall: Each quarter Class A is paid first, then Class B, and the residual goes to the bank. No carry-forwards: a shortfall is not made up later and no cash is held back.

Units: \$ millions. Totals are undiscounted sums over the 20 quarters. The 9% YTM and 1% risk-free rate are not used until Part 2.

2. Implementation Steps

- BIS Bond Function:** Given the period a bond defaults in, the function returns the promised cash flows before it and $(1 - \text{LGD}) \times$ promised from it onward. We checked it against the Lecture 5 slide (face 1000, 5.5%, LGD 60%, π 3%, r 4%) and it gives 984.73, the same as the slide.
- Fixed Random Numbers:** Read the 1000 $\times$ 10 table of independent normals from the csv file (it is created from the seed if it is missing).
- Moment Matching:** Subtract each column's mean, compute the covariance matrix of the draws (dividing by the number of cases, bonds in the order 1 to 10) and its Cholesky factor, and multiply each case's de-meant draws by the inverse of that factor. As drawn, the column means were up to 0.067 away from 0, the variances ran from 0.96 to 1.04 and two bonds' numbers were correlated by up to 0.09. After matching the means are 0, the variances are 1 and the correlations are 0. Only these moments of the normal draws are fixed, so default rates and cash flows are still estimates.
- Correlated Defaults:** Multiply the matched numbers by the Cholesky factor of the correlation matrix, convert to uniforms, then to default times and default quarters. Because of the moment matching the correlated normals have a pairwise correlation of exactly 0.20 across the cases. For the default times it comes out at 0.175.
- Collateral Cash Flows:** Apply the BIS function to every case and bond (1000 $\times$ 10 $\times$ 20 array) and add up the bonds to get the pool.
- Check:** We compare the simulated mean pool cash flow in each quarter with the exact expected value, promised $\times [0.4 + 0.6 \times 0.96^{(k/4)}]$. The largest gap over the 20 quarters is 0.20%.

7. **Waterfall:** $A = \min(\text{pool}, \text{due to A})$, $B = \min(\text{pool} - A, \text{due to B})$, equity = pool - A - B, quarter by quarter. The three always add back to the pool.
8. **Case Viewer:** Setting CASE in the notebook, the case cell in the Excel workbook or the case box on the dashboard (1 to 1000) shows that case's random numbers, default quarters and quarterly cash flows.
9. **Statistics and Sensitivities:** Distributions of total and quarterly cash flows. Then we rerun the model on the same random numbers with the default probability, LGD, correlation and Class B size changed one at a time.
10. **Cross-checks:** We also built the model with live formulas in Excel and in JavaScript for the dashboard, and wrote a plain loop version for the tests. All three give the same cash flows as the Python model in all 1000 cases.

3. Results Summary

Across the 1000 cases an average of 1.82 of the 10 bonds default within 5 years (theory 1.85), and 25.7% of cases have no default at all. The pool collects \$117.3 MM on average out of the \$130 MM promised (90.2%; the exact expected value is \$117.1 MM), with a standard deviation of \$12.0 MM, a 5th percentile of \$94.7 MM and a worst case of \$64.5 MM (case 760, 9 defaults). Class A and Class B are paid in full in every case. This is not luck in the sample: with a 60% LGD the pool can never pay less than 40% of what it promised, which is \$0.60 MM per quarter and \$40.60 MM at maturity, and the two classes together need only \$0.20 MM and \$30.20 MM. Under the BIS model with these inputs both classes are effectively free of default risk, so all of the variability in the collateral passes to the bank's equity. Equity receives \$83.3 MM on average against \$96 MM if nothing defaults, with the same standard deviation as the pool (\$12.0 MM), a 5th percentile of \$60.7 MM and a minimum of \$30.5 MM. Correlation matters for the tails: 15.9% of cases have 4 or more defaults (16.3% exactly under the model), against 9.5% if the bonds defaulted independently.

	No-default amount	Mean	Std error of the mean	Std dev	5th pct	Median	Worst case	Mean / no-default amount
Collateral pool	130.00	117.28	0.38	12.03	94.69	122.29	64.48	90.2%
Class A	22.00	22.00	0.00	0.00	22.00	22.00	22.00	100.0%
Class B	12.00	12.00	0.00	0.00	12.00	12.00	12.00	100.0%
Equity (bank)	96.00	83.28	0.38	12.03	60.69	88.29	30.48	86.8%

Table 1. Total cash received over the 5 years by the collateral pool and by each class, \$ MM, undiscounted, across the 1000 cases. The no-default amount is what each would receive if no bond defaulted. The standard error is the standard deviation divided by the square root of the number of cases, the usual formula for independent cases. Moment matching ties the cases together, so the formula overstates the error of the pool and equity means: across 2,000 fresh sets of random numbers the mean pool cash had a standard deviation of \$0.15 MM with moment matching and \$0.39 MM without.

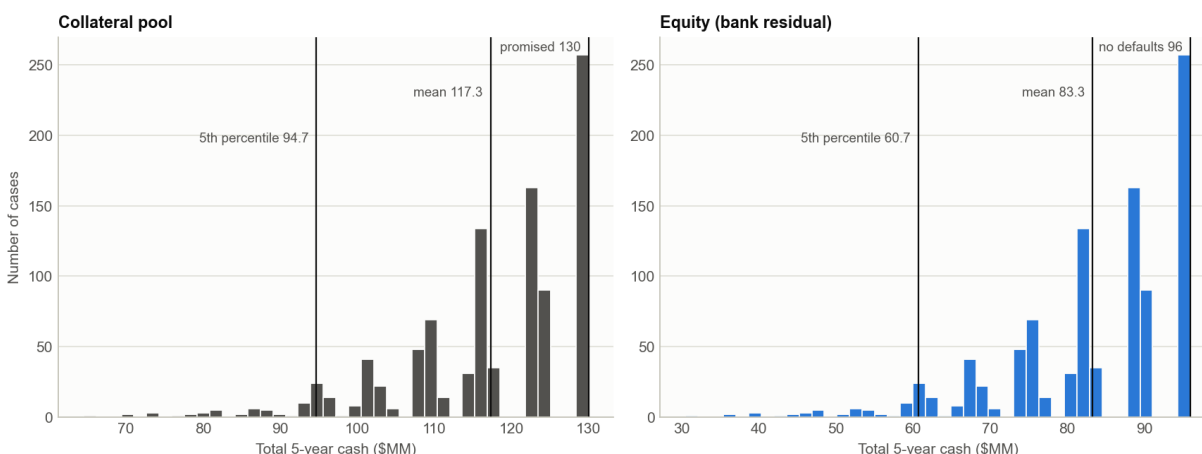


Figure 1. Distribution of total 5-year cash from the collateral pool (left) and to the bank's equity (right). The equity distribution is the pool distribution shifted down by the \$34 MM paid to Classes A and B.

Defaults in 5 years	Cases	Simulated	Exact (model)	If independent	Avg pool cash	Avg equity cash
0	257	25.7%	24.70%	12.99%	130.00	96.00
1	253	25.3%	25.88%	29.41%	123.13	89.13
2	200	20.0%	19.88%	29.97%	116.09	82.09
3	131	13.1%	13.28%	18.10%	109.13	75.13
4	77	7.7%	8.05%	7.17%	101.91	67.91
5	49	4.9%	4.46%	1.95%	94.97	60.97
6	16	1.6%	2.25%	0.37%	87.54	53.54
7	11	1.1%	1.00%	0.05%	79.94	45.94
8	5	0.5%	0.38%	0.00%	72.50	38.50
9	1	0.1%	0.11%	0.00%	64.48	30.48
10	0	0.0%	0.02%	0.00%	-	-

Table 2. Number of bonds defaulting within 5 years: share of the simulated cases, the exact probability under the model (common factor, no random numbers) and the binomial if defaults were independent, with the average cash (\$ MM) in cases with that many defaults. Each default costs the pool about \$7 MM.

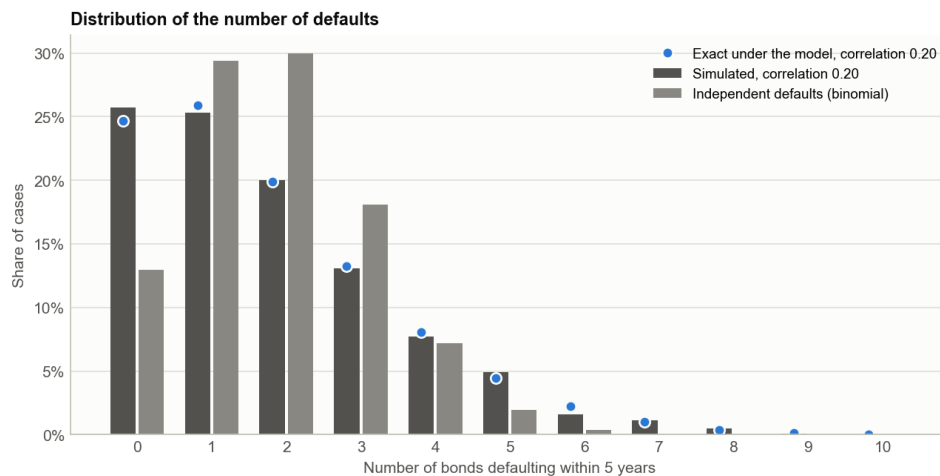


Figure 2. Distribution of the number of defaults. With correlation 0.20 both ends are more likely than under independence: more cases with no defaults and more cases with 5 or more. The dots are the exact probabilities under the model, which the 1000 cases track closely.

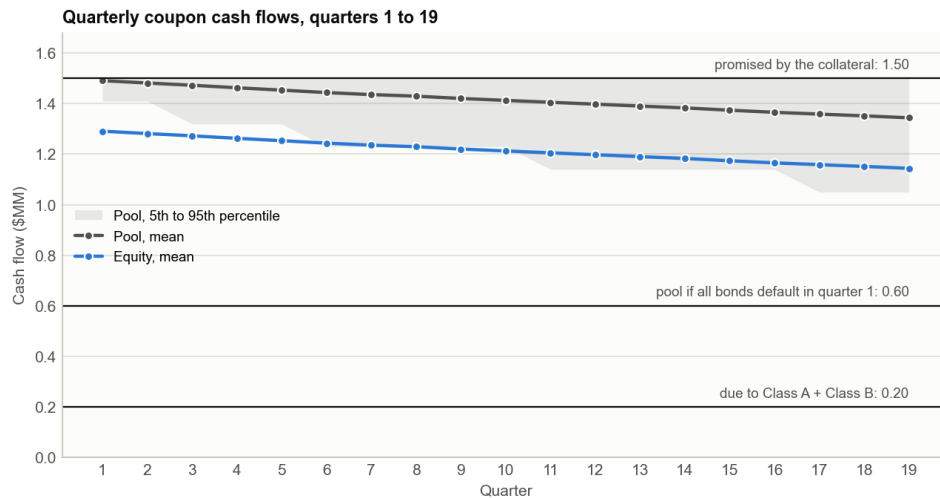


Figure 3. Quarterly coupon cash flows in quarters 1 to 19. The mean pool cash flow falls from \$1.49 MM to \$1.34 MM as defaults build up, and equity receives the pool cash less the \$0.20 MM due to the classes. Even if every bond defaulted in quarter 1 the pool would pay 3 times what the classes need. At maturity (not shown) the pool pays \$90.4 MM on average out of \$101.5 MM, of which equity receives \$60.2 MM.

4. Sensitivities

We changed each input on its own, keeping the others at their base values and using the same fixed random numbers. Because the classes are fully covered in the base case, the sensitivities show up in the equity. Pool cash is equity cash plus what the two classes are paid, so while both classes are paid in full the pool figures are the equity figures plus \$34 MM, with the same standard deviation.

Default probability: the largest effect on the average. Going from 1% to 12% per year takes mean equity cash from \$92.4 MM to \$62.7 MM and the 5th percentile from \$81.5 MM to \$33.6 MM. Around the base case each extra 1% of annual default probability costs equity roughly \$2.9 MM.

LGD: the loss scales in a straight line. Every 10 points of LGD lowers mean equity cash by \$2.1 MM and the 5th percentile by \$5.9 MM. LGD is also the input that decides whether the classes are at risk. Class B is fully covered as long as the worst case pool at maturity, $101.5 \times (1 - \text{LGD})$, is at least 30.2, that is up to an LGD of 70.2%, and Class A up to 80.2%. In the simulation Class B shortfalls first appear at LGD 80% (0.1% of cases) and stand at 0.6% at LGD 90% and 0.6% at LGD 100%.

Correlation: it does not change the average (mean equity stays near \$83 MM, since each bond's own default probability is unchanged) but it widens the distribution. From a correlation of 0 to 0.8 the standard deviation of equity cash rises from \$8.8 MM to \$21.3 MM and the 5th percentile falls from \$67.7 MM to \$28.9 MM, while the chance of no defaults at all rises from 14.5% to 59.4%.

Input	Value	Avg defaults	P(no default)	Equity mean	Equity std	Equity 5th pct	Equity min	P(B short)
Default prob.	1%	0.51	65.2%	92.44	6.01	81.48	46.98	0.0%
Default prob.	2%	0.96	47.2%	89.30	8.56	73.59	44.01	0.0%
Default prob.	4% (base)	1.82	25.7%	83.28	12.03	60.69	30.48	0.0%
Default prob.	6%	2.64	13.6%	77.51	14.25	50.96	28.86	0.0%
Default prob.	8%	3.43	7.6%	72.02	15.60	44.55	21.87	0.0%
Default prob.	12%	4.73	1.9%	62.71	16.56	33.61	20.34	0.0%
LGD	40%	1.82	25.7%	87.52	8.02	72.46	52.32	0.0%
LGD	50%	1.82	25.7%	85.40	10.03	66.57	41.40	0.0%
LGD	60% (base)	1.82	25.7%	83.28	12.03	60.69	30.48	0.0%
LGD	70%	1.82	25.7%	81.16	14.04	54.80	19.56	0.0%
LGD	80%	1.82	25.7%	79.05	16.04	48.91	10.42	0.1%
LGD	90%	1.82	25.7%	76.94	17.97	43.03	8.63	0.6%
LGD	100%	1.82	25.7%	74.88	19.80	37.14	6.90	0.6%
Correlation	0.0	1.83	14.5%	83.22	8.76	67.68	45.54	0.0%
Correlation	0.1	1.82	20.7%	83.28	10.57	63.38	44.01	0.0%
Correlation	0.2 (base)	1.82	25.7%	83.28	12.03	60.69	30.48	0.0%
Correlation	0.4	1.84	35.7%	83.16	15.38	52.62	19.98	0.0%
Correlation	0.6	1.82	46.3%	83.33	17.99	40.98	18.27	0.0%
Correlation	0.8	1.78	59.4%	83.55	21.29	28.87	18.00	0.0%

Table 3. One-at-a-time sensitivities on the same 1000 cases. Equity figures are total 5-year cash in \$ MM. P(B short) is the share of cases in which Class B is paid less than it is due in any quarter.

Sensitivity of the equity cash flows (other inputs at base values)

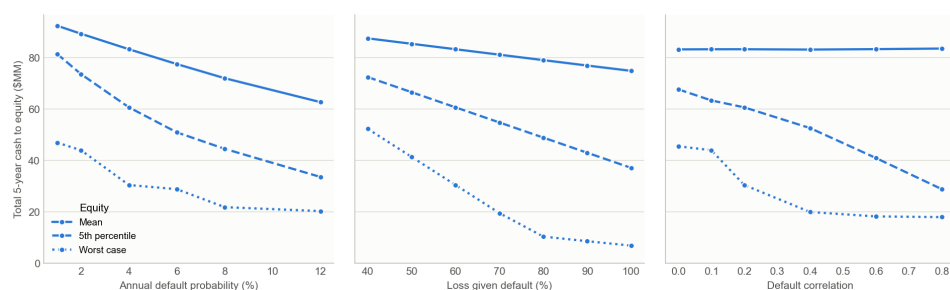


Figure 4. Mean, 5th percentile and worst case of total equity cash as each input is varied.

5. When do the classes become risky?

With the base LGD no combination of default probability and correlation can touch Class A or Class B, so for the runs below we set the LGD to 100% (nothing recovered on a defaulted bond). Each surviving bond pays \$10.15 MM at maturity, so Class B needs at least 3 of the 10 bonds to survive to be paid in full and Class A needs at least 2. At the base default probability and correlation this fails for Class B in only 0.6% of cases, but it rises quickly with either input: to 11.8% at a 12% default probability and to 8.9% at a correlation of 0.8. At the base default probability with independent defaults there are no shortfalls in any of the 1000 cases, so at that default rate the risk to the classes comes from the correlation. These are tail events, so 1000 cases measure them roughly: the base row rests on 6 cases for Class B and 1 for Class A, and the exact probabilities under the model are 0.50% and 0.13%.

Input	Value	Avg defaults	Equity mean	Equity 5th pct	P(A shortfall)	P(B shortfall)	B paid / due
Default prob.	4%	1.82	74.88	37.14	0.1%	0.6%	99.5%
Default prob.	8%	3.43	56.69	11.80	1.9%	4.5%	96.2%
Default prob.	12%	4.73	42.42	8.20	5.6%	11.8%	90.0%
Default prob.	20%	6.73	23.75	4.65	24.7%	40.6%	65.4%
Correlation	0.0	1.83	74.71	48.80	0.0%	0.0%	100.0%
Correlation	0.2	1.82	74.88	37.14	0.1%	0.6%	99.5%
Correlation	0.4	1.84	75.08	23.69	1.6%	2.9%	97.5%
Correlation	0.6	1.82	75.87	12.04	3.2%	5.1%	95.6%
Correlation	0.8	1.78	77.42	8.35	6.9%	8.9%	92.1%

Table 4. Stress runs with LGD = 100%. The default probability rows use correlation 0.20 and the correlation rows use a 4% default probability.

The size of Class B is the other lever. In the worst case the pool pays \$40.6 MM at maturity and Class A takes \$20.1 MM, so Class B stays fully covered by recoveries alone up to a notional of about \$20.3 MM, 2.0 times its proposed size. At \$30 MM it is short in 0.1% of cases and at \$60 MM in 15.9%. Each extra \$1 MM of Class B takes about \$1.20 MM out of the equity cash (its principal plus five years of coupons).

Class B notional	Equity mean	Equity 5th pct	Equity min	P(A shortfall)	P(B shortfall)	B paid / due
\$10 MM (base)	83.28	60.69	30.48	0.0%	0.0%	100.00%
\$20 MM	71.28	48.69	18.48	0.0%	0.0%	100.00%
\$30 MM	59.29	36.69	10.19	0.0%	0.1%	99.99%
\$40 MM	47.35	24.69	8.20	0.0%	1.7%	99.85%
\$50 MM	35.62	12.78	6.30	0.0%	3.3%	99.45%
\$60 MM	24.70	10.07	4.40	0.0%	15.9%	98.04%

Table 5. Class B notional varied with all other inputs at base (LGD 60%). Equity figures are total 5-year cash in \$ MM.

6. Points for the client

- Class A and Class B are covered by the recovery value of the collateral alone. As long as recoveries are at least 30% of promised payments (LGD of 70% or less), they are paid in full whatever the default experience.
- The bank's retained equity carries all of the default risk. Its expected cash is about 87% of the no-default amount, and in 1 case out of 20 it receives \$61 MM or less out of \$96 MM.
- Over the ranges we tested, the default probability moves the equity result the most. Correlation leaves the average alone but lowers the 5th percentile from \$68 MM to \$29 MM between 0 and 0.8. The LGD assumption is the one to watch for the classes.
- The structure is conservative: Class B could be about \$20 MM instead of \$10 MM and still be fully covered if every bond defaulted.
- These are cash flow results only. What the classes and the equity are worth, given the 9% market yield on the collateral and the 1% risk-free rate, is the subject of Part 2.

Appendix A. Quarterly cash flows across the 1000 cases (\$ MM)

Quarter	Pool promised	Pool exp. (exact)	Pool mean	Pool 5th pct	Pool 95th pct	Class A	Class B	Equity mean	Equity 5th pct
1	1.50	1.491	1.491	1.41	1.50	0.10	0.10	1.291	1.21
2	1.50	1.482	1.482	1.41	1.50	0.10	0.10	1.282	1.21
3	1.50	1.473	1.473	1.32	1.50	0.10	0.10	1.273	1.12
4	1.50	1.464	1.463	1.32	1.50	0.10	0.10	1.263	1.12
5	1.50	1.455	1.453	1.32	1.50	0.10	0.10	1.253	1.12
6	1.50	1.447	1.444	1.23	1.50	0.10	0.10	1.244	1.03
7	1.50	1.438	1.436	1.23	1.50	0.10	0.10	1.236	1.03
8	1.50	1.429	1.430	1.23	1.50	0.10	0.10	1.230	1.03
9	1.50	1.421	1.420	1.23	1.50	0.10	0.10	1.220	1.03
10	1.50	1.413	1.413	1.23	1.50	0.10	0.10	1.213	1.03
11	1.50	1.404	1.405	1.14	1.50	0.10	0.10	1.205	0.94
12	1.50	1.396	1.398	1.14	1.50	0.10	0.10	1.198	0.94
13	1.50	1.388	1.390	1.14	1.50	0.10	0.10	1.190	0.94
14	1.50	1.380	1.383	1.14	1.50	0.10	0.10	1.183	0.94
15	1.50	1.372	1.374	1.14	1.50	0.10	0.10	1.174	0.94
16	1.50	1.364	1.366	1.14	1.50	0.10	0.10	1.166	0.94
17	1.50	1.357	1.359	1.05	1.50	0.10	0.10	1.159	0.85
18	1.50	1.349	1.352	1.05	1.50	0.10	0.10	1.152	0.85
19	1.50	1.341	1.344	1.05	1.50	0.10	0.10	1.144	0.85
20	101.50	90.256	90.410	71.05	101.50	20.10	10.10	60.210	40.85

Class A and Class B receive the same amount in every case, so only one column is shown for each. The pipeline also writes this table to output/quarterly_cash_flow_summary.csv.

Appendix B. Example of a single case (case 5)

In case 5, 6 bonds default: bond 1 in quarter 4, bond 4 in quarter 20, bond 5 in quarter 5, bond 6 in quarter 13, bond 7 in quarter 12, bond 9 in quarter 7. The pool, class and equity cash flows are below. Any other case can be shown by changing CASE in the notebook, the case cell in the Excel workbook or the case box on the dashboard.

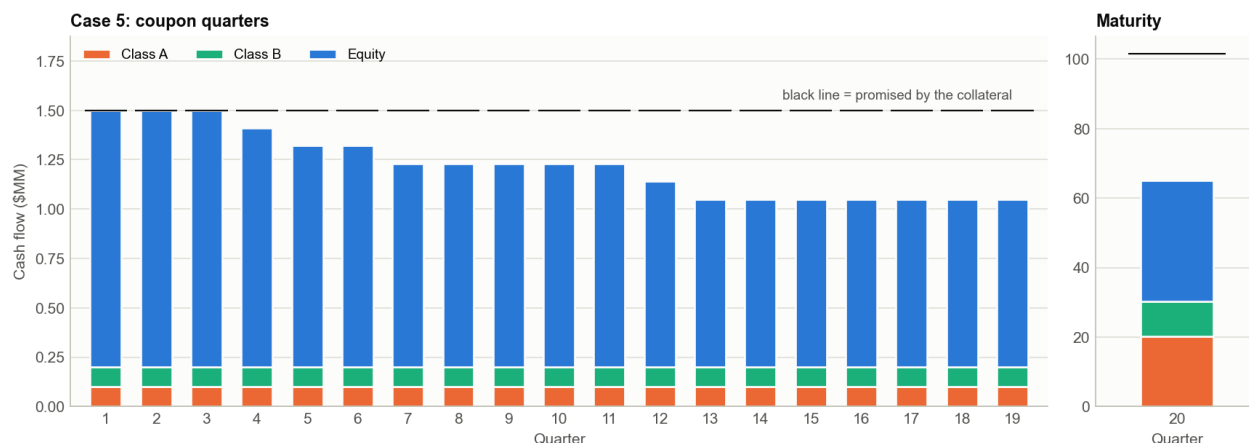


Figure 5. Case 5: who receives the pool's cash each quarter. The classes take the same amount every quarter and the equity absorbs each default as it happens.

Quarter	Bonds in default	Pool	Class A	Class B	Equity
1	0	1.50	0.10	0.10	1.30
2	0	1.50	0.10	0.10	1.30
3	0	1.50	0.10	0.10	1.30
4	1	1.41	0.10	0.10	1.21
5	2	1.32	0.10	0.10	1.12
6	2	1.32	0.10	0.10	1.12
7	3	1.23	0.10	0.10	1.03
8	3	1.23	0.10	0.10	1.03
9	3	1.23	0.10	0.10	1.03
10	3	1.23	0.10	0.10	1.03
11	3	1.23	0.10	0.10	1.03
12	4	1.14	0.10	0.10	0.94
13	5	1.05	0.10	0.10	0.85
14	5	1.05	0.10	0.10	0.85
15	5	1.05	0.10	0.10	0.85
16	5	1.05	0.10	0.10	0.85
17	5	1.05	0.10	0.10	0.85
18	5	1.05	0.10	0.10	0.85
19	5	1.05	0.10	0.10	0.85
20	6	64.96	20.10	10.10	34.76
Total		88.15	22.00	12.00	54.15